



# Interaksi Manusia dan Komputer (IMK) THE USER INTERFACE (UI) DESIGN PROCESS

Step - 8  
Clear Text and Messages

Step - 9  
Provide Effective Feedback  
and Guidance and Assistance

# Interaksi Manusia dan Komputer (IMK)

## THE USER INTERFACE (UI) DESIGN PROCESS

Step - 8  
Clear Text and Messages



# Tujuan

Setelah mengikuti materi ini mahasiswa dapat:

- Mahasiswa/i mampu merancang teks dan pesan yang tepat dan sesuai dengan prinsip UI

# Outline (Step 8)

- Panduan dalam penulisan teks dan pesan
  - General :
    - The concept of readability.
    - Choosing the proper words.
    - Writing sentences and messages.
    - Kinds of messages.
    - Presenting and writing text.
    - Window title, conventions, and sequence control guidance
  - Web Spesific Guidelines :
    - Presenting and writing page text.
    - Writing links and headings.
    - Writing instructions and error messages

# Outline (Step 8)

- Contoh penulisan teks
- Feedback
- Guidance & Assistance
- Contoh kasus feedback
- Contoh kasus Guidance dan Assistance

# Outline (Step 9)

- Feedback
- Guidance & Assistance
- Contoh kasus Feedback
- Contoh kasus Guidance & Assistance

# Panduan dalam penulisan teks dan pesan

# Readability

- The degree to which prose can be **understood**, based upon the complexity of its words and sentences
- Some formula to measure readability :
  - the *Automated Readability Index*
  - the *Gunning-Fog Index* and
  - the *Dale-Chall-Flesch Reading Ease*
  - *Flesch-Kincaid Grade Level*

## Readability Consensus

Based on 8 readability formulas, we have scored your text:

Grade Level: 7

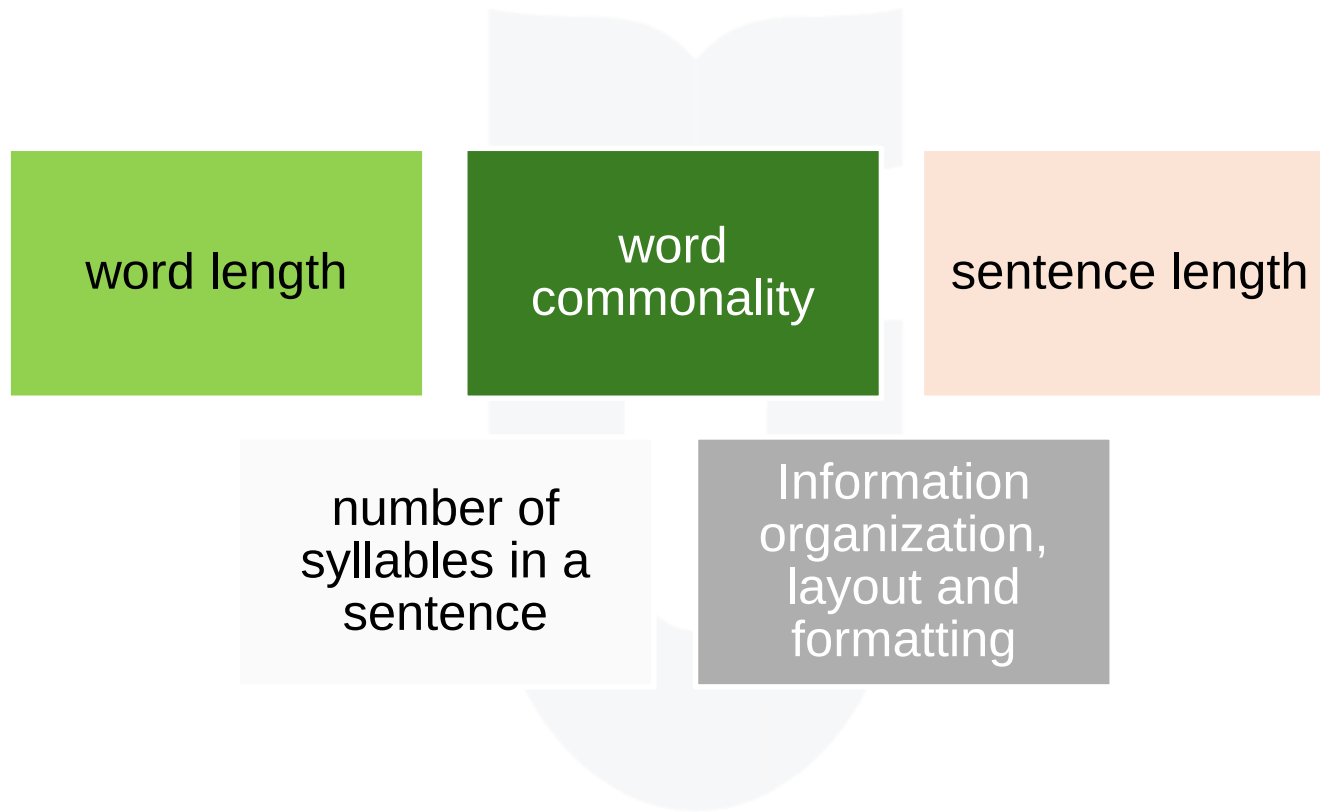
Reading Level: standard / average.

Reader's Age: 11-13 yrs. old (Sixth and Seventh graders)

- Most reading formulas include only two factors:
  - The number of **syllables** or **(letters)** in a word
  - The number of **words** in a sentence



# Factors Influence Readability



# Readability Guidelines



Write at a **readability** level below the reading skill level of the intended audience.



Apply all the other principles for clear writing and text presentation

# Choosing The Proper Words

## Do not use

- Jargon, words, or terms
- Abbreviations or acronyms
- Word contractions, suffixes, and prefixes

## Use

- Short, familiar words.
- Standard alphabetic characters.
- Complete words.
- Positive terms.
- Simple action words; avoid noun strings.
- The “more” dimension when comparing.
- Consistent words
- Punctuation for abbreviations, mnemonics, and acronyms

## Do Not

- Stack Words
- Hyphenate words

# Some Words to Forget

| AVOID      | USE                              |
|------------|----------------------------------|
| Abend      | End, Cancel, Stop                |
| Abort      | End, Cancel, Stop                |
| Access     | Get, Ready, Display              |
| Available  | Ready                            |
| Boot       | Start, Run                       |
| Error      | —                                |
| Execute    | Complete                         |
| Hit        | Press, Depress                   |
| Implement  | Do, Use, Put Into                |
| Invalid    | Not Correct, Not Good, Not Valid |
| Key        | Type, Enter                      |
| Kill       | End, Cancel                      |
| Output     | Report, List, Display            |
| Return Key | Enter, Transmit                  |
| Terminate  | End, Exit                        |

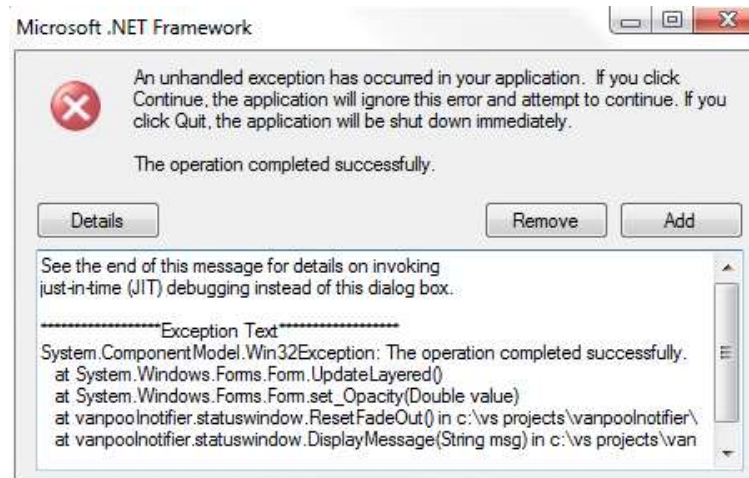
# Writing Sentences & Messages

- Sentences and messages must be
  - Brief and simple.
  - Limited to no more than **twenty words** per sentence.
  - Limited to no more than **six sentences** per paragraph.
  - Written at an **eighth grade level or less** for the general population.
  - Directly and immediately usable.
  - An **affirmative** statement.
  - In an **active** voice.
  - In the temporal sequence of events
  - Structured so that the main topic is near the beginning
  - Of **parallel** construction

# Writing Sentences & Messages

- Sentences and messages must be of the **proper tone**:
  - Non authoritarian
  - Non threatening
  - Non anthropomorphic
  - Non patronizing
  - Non punishing
  - Cautious in the use of humor.

# Kinds of Messages



- Messages are communications provided on the screen to the screen viewer
- Screen messages categories:
  - System Messages
    - generated by the system to **keep the user informed** of the system's state and activities
    - They are customarily presented within message boxes
  - Instructional Messages
    - Sometimes referred to as *prompting messages*
    - Messages that tell the user **how to work with**, or complete, the screen displayed
    - They may be provided in messages boxes and also within the screen itself.

# System Messages Types

**Status**—Provides information concerning the progress of a lengthy operation

**Informational**—Provides information about the state of the system when it is not immediately obvious

**Warning**—Calls attention to a situation that may be undesirable

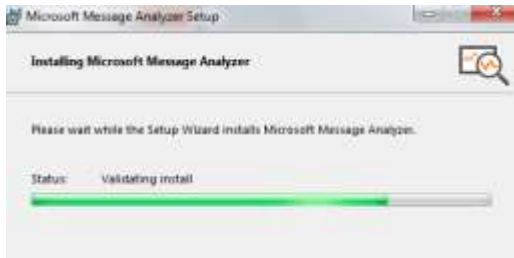
**Critical**—Calls attention to conditions that require a user action before the system can proceed

**Question**—Asks a question and offers options

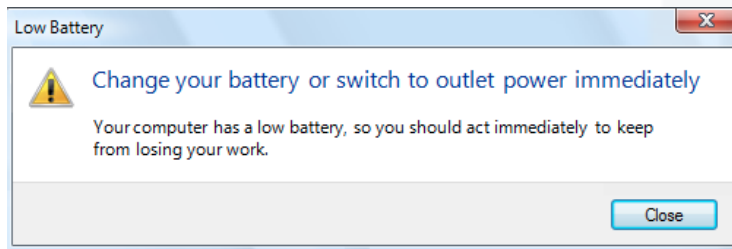


# System Messages Example

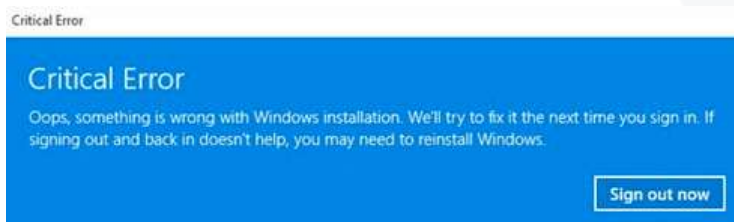
- **Status**



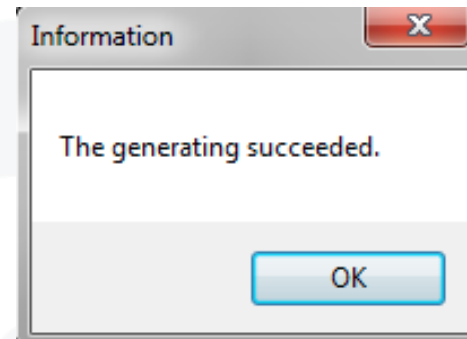
- **Warning**



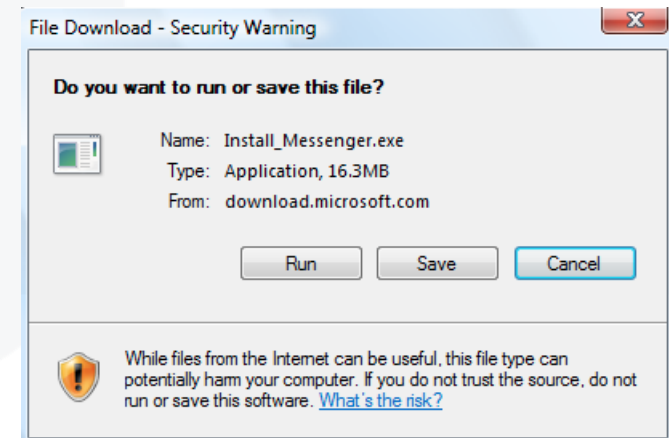
- **Critical**



- **Informational**



- **Question**



# Writing Message Box Text

- Title bar:
  - Clearly identify the source of the message
    - The name of the object to which it refers
    - The name of the application to which it refers.
  - Do not include an indication of message type.
  - Use mixed case in the headline style.
- Message box:
  - Provide a clear and concise description of the condition causing the message box to be displayed
    - Use complete sentences with ending punctuation
    - State the problem, its probable cause (if known), and what the user can do about it
    - Avoid contractions & technical jargon

# Writing Message Box Text

- Provide only as much background information as necessary for the message to be understood
- Show only one message box about the cause of condition in a single message
- Make the solution an option offered in the message
- Avoid multistep solutions
- Use consistent words and phrasing for similar situations
- Use the word *please conservatively*
- Do not exceed two or three lines
- Include the relevant icon identifying the type of message to the left of the text
- Center the message text in window

# Message Box Controls

- Command buttons:
  - If a message requires no choices to be made but only acknowledgment
    - Include an OK button
  - If a message requires a choice to be made, provide a command button for each option
    - Include OK and Cancel buttons only when the user has the option of continuing or stopping the action
    - Include Yes and No buttons when the user must decide how to continue
    - If these choices are too ambiguous, label the command buttons with the names of specific actions

# Message Box Controls (cont'd)

- If a message allows initiation of an action to correct the situation described,
  - Include a properly labeled button initiating the corrective action
- If a message describes an interrupted process whose state cannot be restored,
  - Provide a Stop button.
- If a message offers an opportunity to cancel a process as well as to perform or not perform an action
  - Provide a Cancel button
- If more details about a message topic must be presented
  - Provide a Help button

# Message Box Controls (cont'd)

- Designate the most frequent or least destructive option as the default
- Display a message box only when the window of an application is active
- Display only one message box for a specific condition
- Close box:
  - Enable the title bar Close box only if the message includes a Cancel button.

# Message Location

- Use the message line for messages that must not interfere with screen information
- Pop-up windows may be used for all kinds of messages, if available
- Pop-up windows should always be used for critical messages.

# Instructional Messages

- Provide instructional information at the depth of detail needed by the user
- Locate it at strategic positions on the screen
- Display it in a manner that visually differentiates it from other screen elements
- In writing, follow all relevant writing guidelines for words, sentences, and messages



# Presenting Text

- Display prose text in mixed upper- and lower-case letters.
- Fonts:
  - Use plain and simple fonts
  - Choose a minimum point size of 12 to 14.
  - Use proportional fonts
- Justification:
  - Left-justify
  - Do not right-justify
  - Do not hyphenate words
- Line Length:
  - If fast reading is required, use line lengths of about 75-100 characters

# Presenting Text (Cont'd)

- If user preference is important, a length of about 50-60 characters is acceptable
- Very narrow columns should be avoided
- Line endings:
  - Coincide with grammatical boundaries
- Line spacing:
  - Increase line spacing to enhance legibility and readability
- Content:
  - Use headings to introduce a new topic
  - Make first sentences descriptive Separate paragraphs by at least one blank line
  - Start a fresh topic on a new page
  - Use lists to present facts

# Presenting Text (Cont'd)

- Emphasize important things by
  - Positioning
  - Boxes
  - Bold typefaces
  - Indented margins
- Provide a screen design philosophy consistent with other parts of the system

# Writing Text

## Sentences and paragraphs

- Use short sentences
- Cut the number of sentences
- Use separate sentences for separate ideas
- Keep the paragraphs short
- Restrict a paragraph to only one idea

## Style

- Use the active writing style
- Use the personal writing style
- Write as you talk
- Use subjective opinion
- Use specific examples
- Read it out loud

# Window title, Conventions, & Sequence Control Guidance

- ***Window Title***

- All windows must have a title located in a centered position at the top
  - Exception: Windows containing messages
- Clearly and concisely describe the purpose of the window
- Spell it out fully using an uppercase or mixed-case font.
- If title truncation is necessary, truncate it from right to left.
- If presented above a menu bar, display it with a background that contrasts with the bar

# Window title, Conventions, & Sequence Control Guidance

- ***Conventions***

Establish conventions for referring to

- Individual keyboard keys
- Keys to be pressed at the same time
- Field captions
- Names supplied by users or defined by the system.
- Commands and actions

# Window title, Conventions, & Sequence Control Guidance

- ***Sequence Control Guidance***
  - Consider providing a guidance message telling how to continue at points in the dialog when
    - A decision must be made
    - A response needs to be made to continue
  - Consider indicating what control options exist at points in the dialog where several alternatives may be available
  - Permit these prompts to be turned on or off by the user

# Presenting & Writing Page Text

- The most important part of a Web site is its content, a well-written site is a necessity
- Poorly written content has been found to increase user confusion, increase comprehension errors, increase reading times, and reduce user satisfaction
- Content re-written properly for presentation on a Web site has been found to reduce the time to perform a task by 80 percent, improve one's memory of the content by 100 percent, and increase user satisfaction by 37 percent (Morkes and Nielsen, 1998)



# Words & Presentation of Page Text

## Words

- Minimize the use of words that call attention to the Web

## Presentation

- Provide text that contrasts highly with the background

# Writing Page Text

- **Style:**
  - Use a style reflecting the needs of the site users
  - Write objectively
  - Use the inverted pyramid organization
  - Be concise, using only about half the number of words of conventional text
  - Each paragraph should
    - Be short
    - Contain only one main idea
- **Links:**
  - Minimize within-text links
  - Place them at the beginning or end of paragraphs or sections of text

# Writing Page Text

- Scanning:
  - Make text scannable by using
    - Bulleted listings
    - Tables
    - Headings and subheadings
    - Highlighted and emphasized important issues
    - Short paragraphs
- International audience:
  - Consider internationalization needs
- Testing:
  - Test for legibility and readability

# Writing Link Labels

- Create meaningful labels that use the following:
  - Descriptive, differentiable, predictive, and active wording
  - Keywords positioned at the beginning
  - Content that is concise but long enough to be understood
  - Wording that clearly indicates link destination or resulting action
  - Link names that match with their destination page
- Integrate embedded links smoothly into the text
- Ensure that embedded links are descriptive
  - Make only a few words the active link
  - Do not spread links over two lines
- Standalone links should not exceed one sentence in length
- Provide glosses or link labels to assist link understanding

# Link Titles

- Provide link titles that describe
  - The name of site the link will lead to (if different from current site).
  - The name of subsection the link will lead to (if staying within current site).
  - The kind of information to be found at the destination.
  - How the linked information relates to the anchor link and the current page content.
  - How large the linked information is.
  - Warnings about possible problems to be encountered at the other end.
- Restrict titles to no more than 60 characters.

# Page Title

- Provide a page title
  - That possesses meaningful keywords
  - Whose first word is its most important descriptor
  - That makes sense when viewed completely out of context
  - That is different from other page titles
  - That is written in mixed case using the headline style
- Do not highlight keywords

# Headings and Headlines

- Create meaningful headings and headlines that quickly communicate the content of what follows
  - Make the first word an important information-carrying one
  - Skip leading articles (the and a)
- Create meaningful subheadings to break up large blocks of text

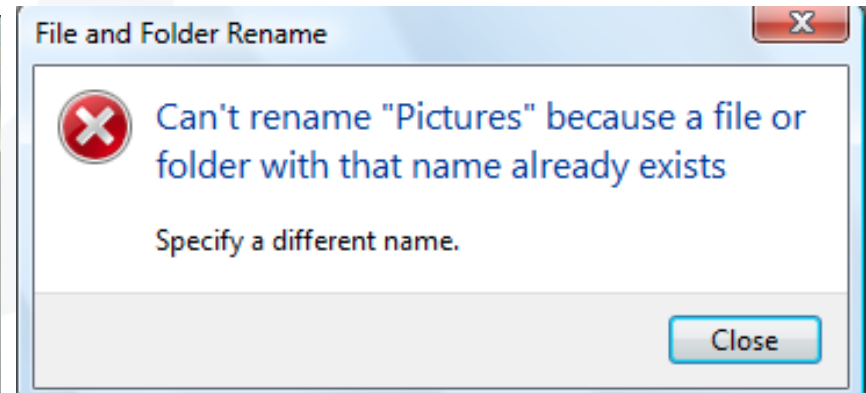
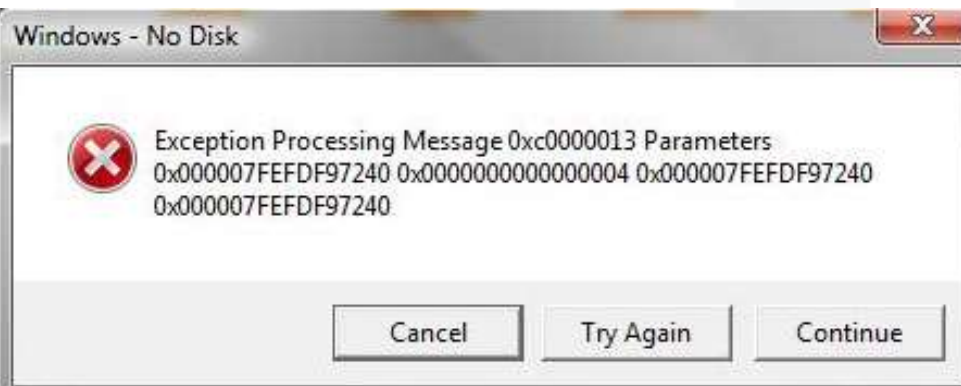
# Instructions

- Do not use phrasing that indicates a certain page order or flow
- Explain where “Up” leads too
- Phrase in a browser-independent manner
- Minimize “Click here” instructions
  - Say “Select this link”



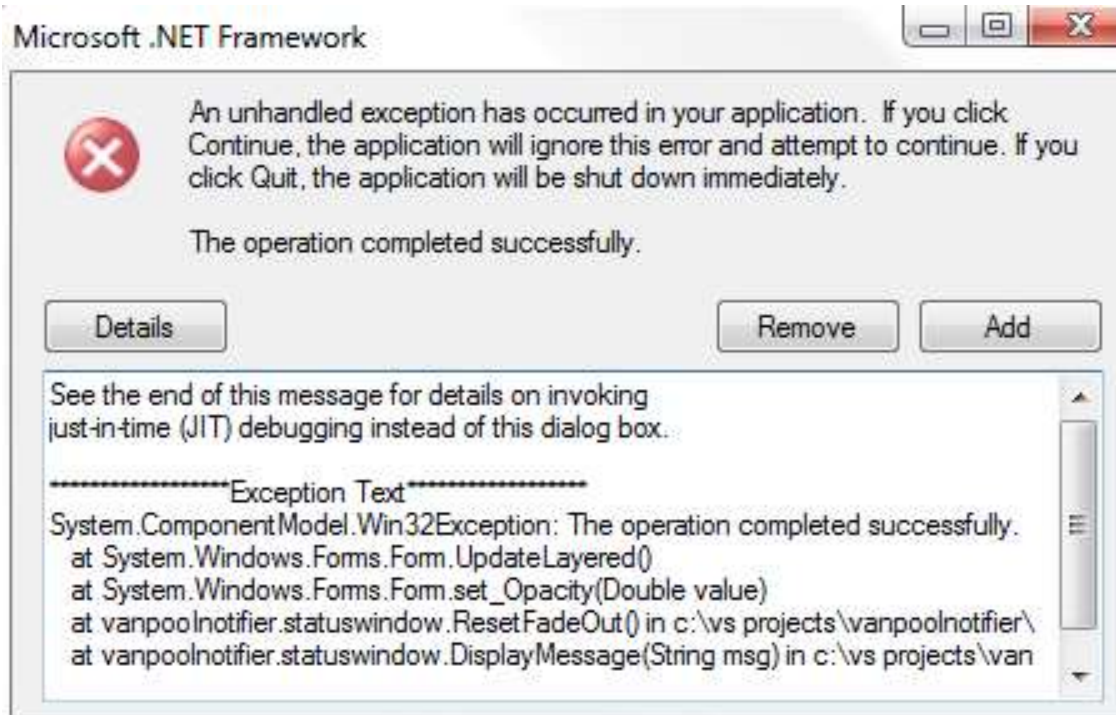
# Error Messages

- Provide helpful and precise error messages for
  - Incomplete or incorrectly keyed, entered, or selected data
  - Requests for documents that do not exist or cannot be found
- Redisplay a message on the page to which it relates
- Present them in a visually distinctive and noticeable manner



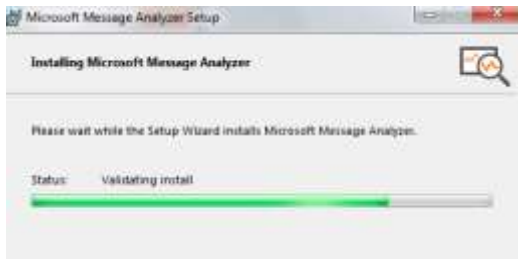
# Contoh kasus penulisan teks

# Instructional Messages

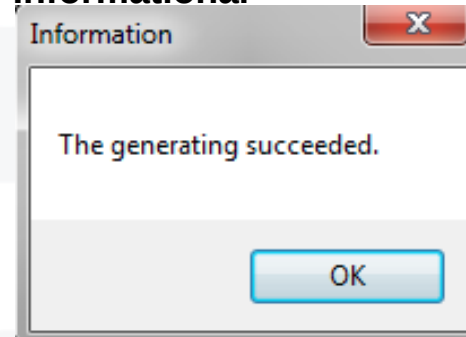


# System Messages Example

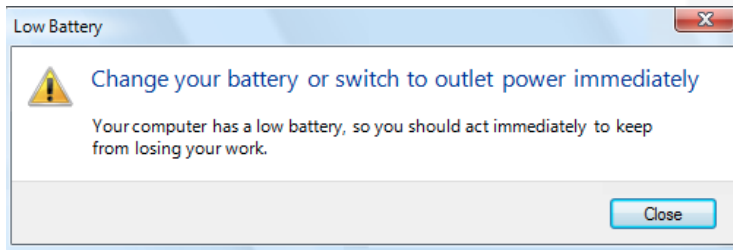
- **Status**



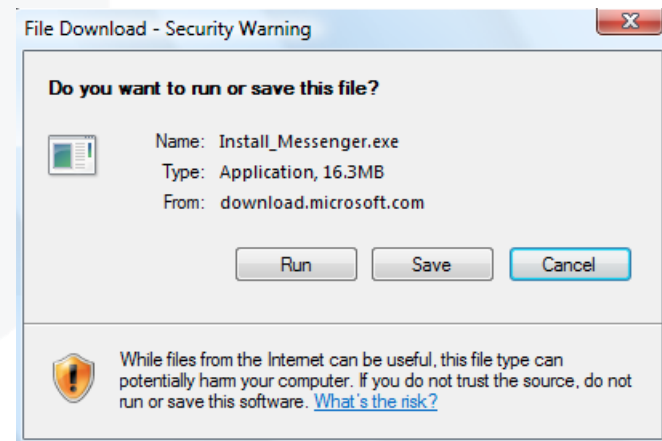
- **Informational**



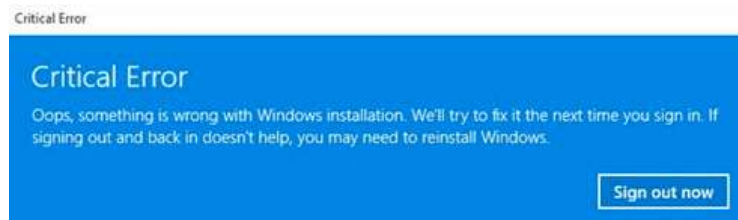
- **Warning**



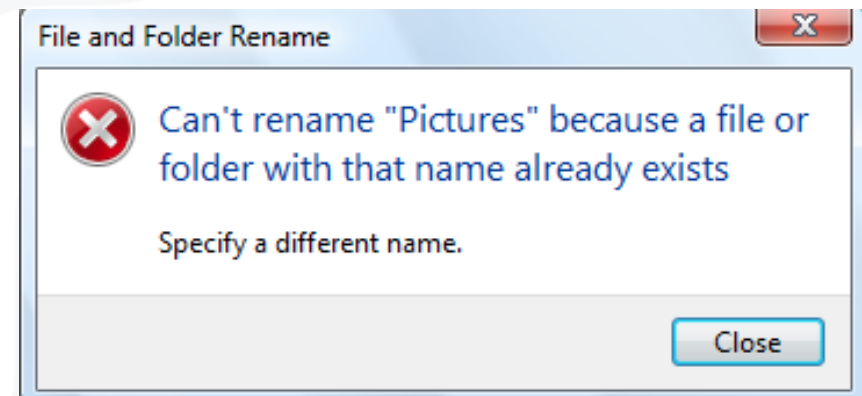
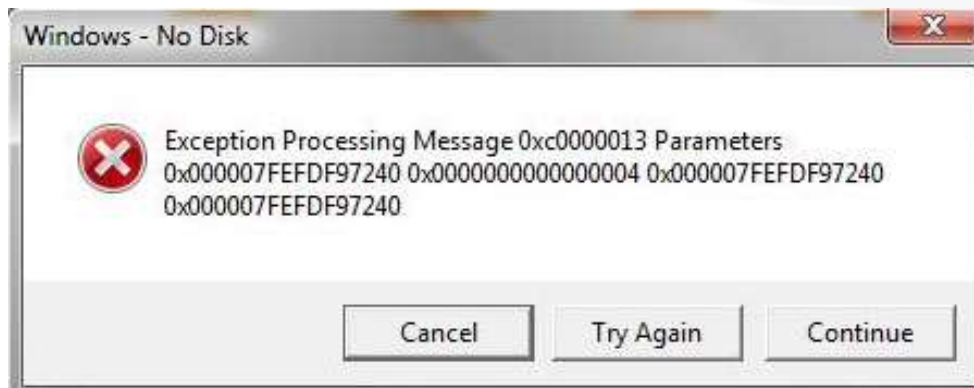
- **Question**



- **Critical**



# Error Messages



# Activities

- Any text/messages/instructional/title/ etc... on your Project?
- Review and revised your project based on the “Step 8” guideline!

# Interaksi Manusia dan Komputer (IMK)

## THE USER INTERFACE (UI) DESIGN PROCESS

Step - 9  
Provide Effective Feedback  
and Guidance and Assistance



# Tujuan

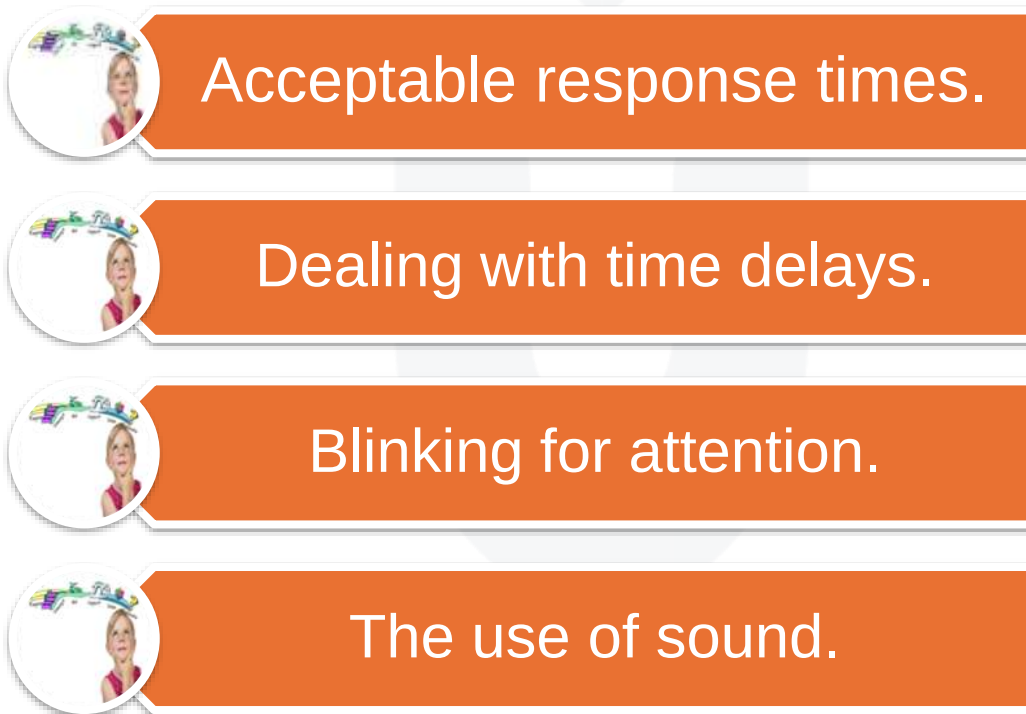
Setelah mengikuti materi ini mahasiswa dapat:

- Mampu merancang *feedback*, *guidance*, dan *assistance* dengan efektif .



# Provide Effective Feedback

- feedback topics are discussed :



# Response times

- The Fact Findings :



The optimum response time is dependent upon the task

Satisfaction with response time is a function of expectations

Dissatisfaction with response time is a function of one's uncertainty about delay

People will change work habits to conform to response time

Constant delays are preferable to variable delays

Very fast or slow response times can lead to symptoms of stress

# Web Page Download Times

- Test participants rated the response times with the following results:
  - High (Good): Up to 5 seconds
  - Average: From 6 to 10 seconds
  - Low (Poor): Over 10 seconds

Other studies have found the following:

- Longer delay times are tolerated by :
  - Novice users, Older users, People not experienced with high bandwidth connections, New visitors to a site, people performing important tasks, People performing successful tasks, Better content
- Exceedingly slow sites can lead people to believe that they made an error

# Fun Fact

When shopping online, what are the biggest irritations and/or what would make you abort?

UK respondents

1,568

|                                                                                                    |       |       |
|----------------------------------------------------------------------------------------------------|-------|-------|
| Slow browsing pages or product images are slow to load                                             | 1,048 | 66.8% |
| It isn't easy to see, e.g. I can't find the product I want; I can't move around the site with ease | 780   | 49.7% |
| Too many steps when I try to purchase goods                                                        | 633   | 40.4% |
| Not being able to gauge size                                                                       | 560   | 35.7% |
| Having to register                                                                                 | 525   | 33.5% |
| Not being able to try on                                                                           | 518   | 33.0% |
| Not being able to get a feel for quality                                                           | 499   | 31.8% |
| Not being able to see true colour                                                                  | 465   | 29.7% |
| Poor design                                                                                        | 461   | 29.4% |
| There is a lack of customer support                                                                | 401   | 25.6% |
| Not being able to touch products                                                                   | 386   | 24.6% |
| Payment process                                                                                    | 306   | 19.5% |
| Design doesn't reflect the brand                                                                   | 153   | 9.8%  |
| Other (please specify)                                                                             | 86    | 5.5%  |

<http://brandperfect.org/>

# Tools for Measurement

PageSpeed Tools > Insights

GUIDES REFERENCE SAMPLES SUPPORT

## PageSpeed Insights

[ANALYZE](#)

Mobile  Desktop

**Good**  
96 / 100

Great job! This page applies most performance best practices and should deliver a good user experience.

 **Possible Optimizations**  
Eliminate render-blocking JavaScript and CSS in above.



 [FULL PAGE TEST](#) [DNS HEALTH](#) [LOG IN](#) [SIGN UP](#)

## Pingdom Website Speed Test

Enter a URL to test the load time of that page, analyze it and find bottlenecks.

URL:  Test from:  [START TEST](#)

Sign up for free to test your site every minute [SIGN UP FOR FREE](#)

Recent tests

| URL                                                                           | TESTED  |
|-------------------------------------------------------------------------------|---------|
| <a href="http://www.szlivingfurniture.ie">http://www.szlivingfurniture.ie</a> | 21s ago |
| <a href="http://www.kreytiv.com">http://www.kreytiv.com</a>                   | 22s ago |

# Web Page Download Times

## Guideline:

- Maximum downloading time should be about 10 seconds.
  - Use incremental or progressive image presentation.
- Images should be
  - Small
  - Lean (Low bit depth, resolution, use of fewer colors, and so forth).
  - Reusable.
  - Contain pixel height and width dimensions in a tag.
- Use thumbnail images to preview larger images.
- Use simple background images

# Dealing with Time Delays

- Button click acknowledgement:
  - Acknowledge all button clicks by visual or aural feedback within one-tenth of a second.
- Waits of up to 10 seconds:
  - If an operation takes 10 seconds or less to complete, present a “busy” signal until the operation is complete.
- Display, for example, an animated hourglass pointer.
- Waits of 10 seconds to 1 minute:
  - If an operation takes longer than 10 seconds to complete, display
  - A rolling barber’s pole or other large animated object.
  - Additionally, a progress indicator, percent complete message, or elapsed time message.

# Dealing with Time Delays

- Waits over 1 minute:
  - Present an estimate of the length of the wait.
  - Display a progress indicator, percent complete message, or elapsed time message.
- Long, invisible operations:
  - When an operation not visible to the user is completed, present an acknowledgement that it is completed.
    - A message Or An auditory tone.
- Progress indicator:
  - Along rectangular bar that is initially empty but filled as the operation proceeds.
    - Dynamically fill the bar.
    - Fill it with a color or shade of gray.
    - Fill it from left to right or bottom to top



# Dealing with Time Delays



**Figure 9.1:** Processing progress indicator.

- **Percent complete message:** Useful if a progress indicator takes too long to update
- **Elapsed time message:** A message that shows the amount of elapsed time
- **Web page downloads**

# Blinking for Attention

- Attract attention by flashing an indicator when an application is inactive but must display a message to the user.
- To provide an additional message indication, also provide an auditory signal (one or two beeps).
- Display the message

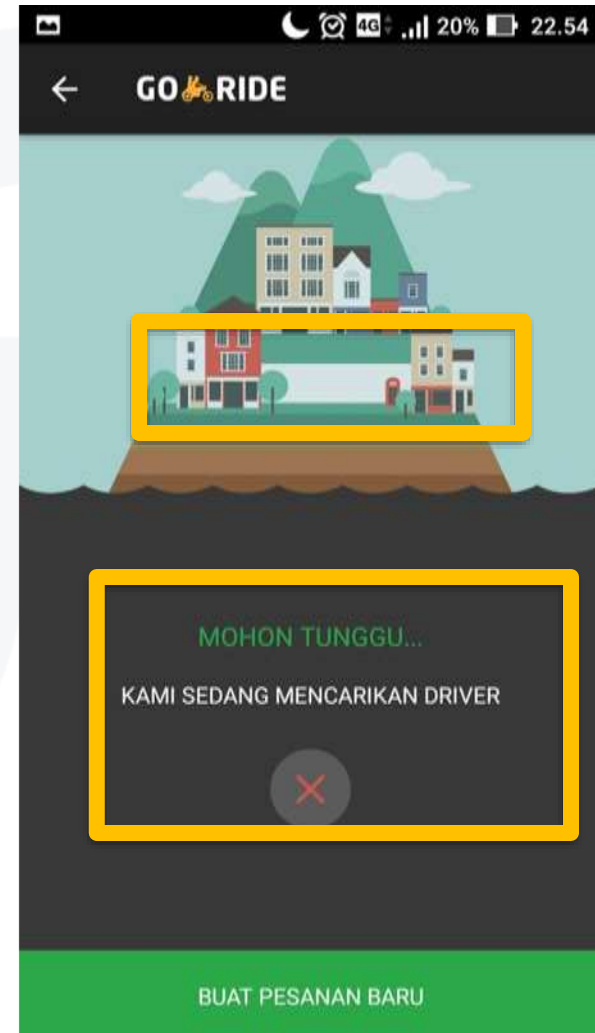


# Use of Sound

- Always use in conjunction with a visual indication.
- Use no more than six different tones.
- Do not use : Jingles or tunes or Loud signals.
- Use tones consistently.
- Provide signal frequencies between 500 and 1,000 Hz.
- Allow the user to adjust the volume or turn the sound off altogether.
- Test the sounds with users over extended trial periods.
- Use sounds sparingly



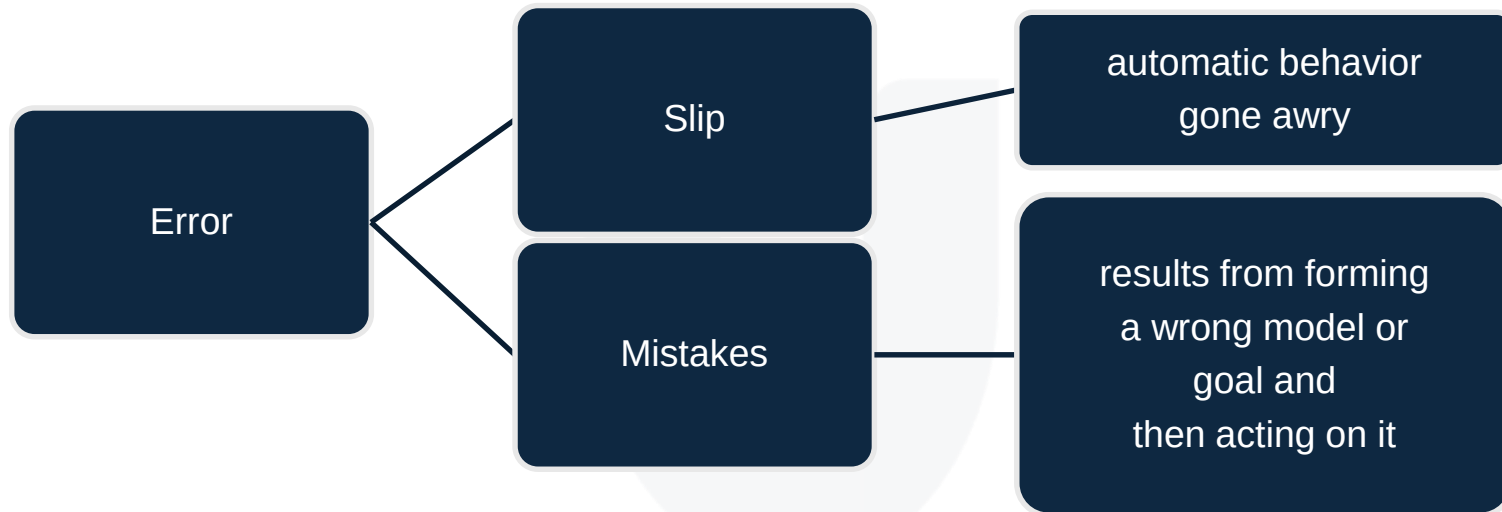
# Contoh studi kasus feedback



# Guidance and Assistance

# Guidance and Assistance

## Preventing Error

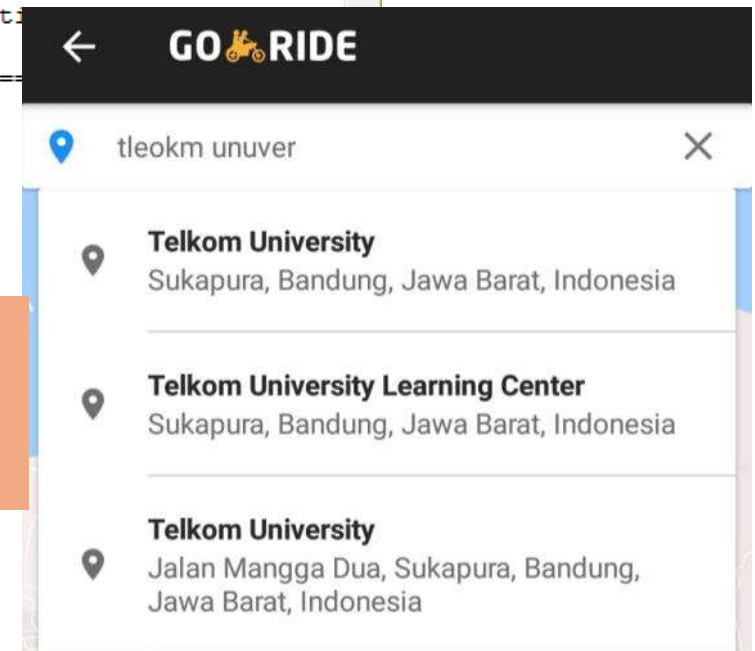


**Everyone makes mistakes, so every mistake should be fixable**

```
Python 3.6.2 Shell
File Edit Shell Debug Options Window Help
Python 3.6.2 (v3.6.2:5fd33b5, Jul 8 2017, 04:14:34) [MSC v.1900 32 bit (Intel)]
on win32
Type "copyright", "credits" or "license()" for more informati
>>>
===== RESTART: D:\session01-a.py =====
Hello world!
Traceback (most recent call last):
  File "D:\session01-a.py", line 2, in <module>
    my_variable == 10
NameError: name 'my_variable' is not defined
>>> |
```

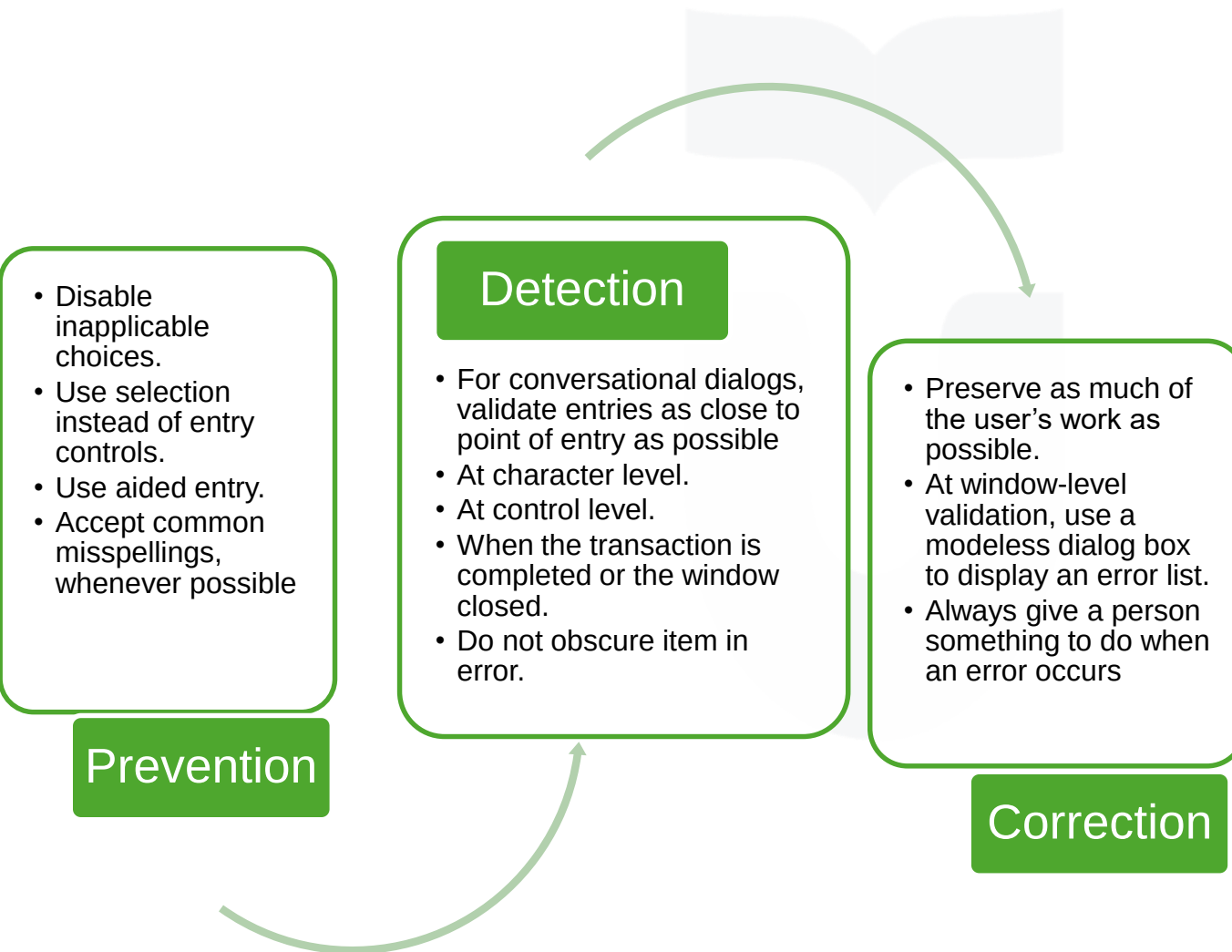
Penyebab  
eror  
difokuskan

Koreksi dan  
rekomendasi  
terhadap salah  
ketik oleh  
pengguna





# Problem Management

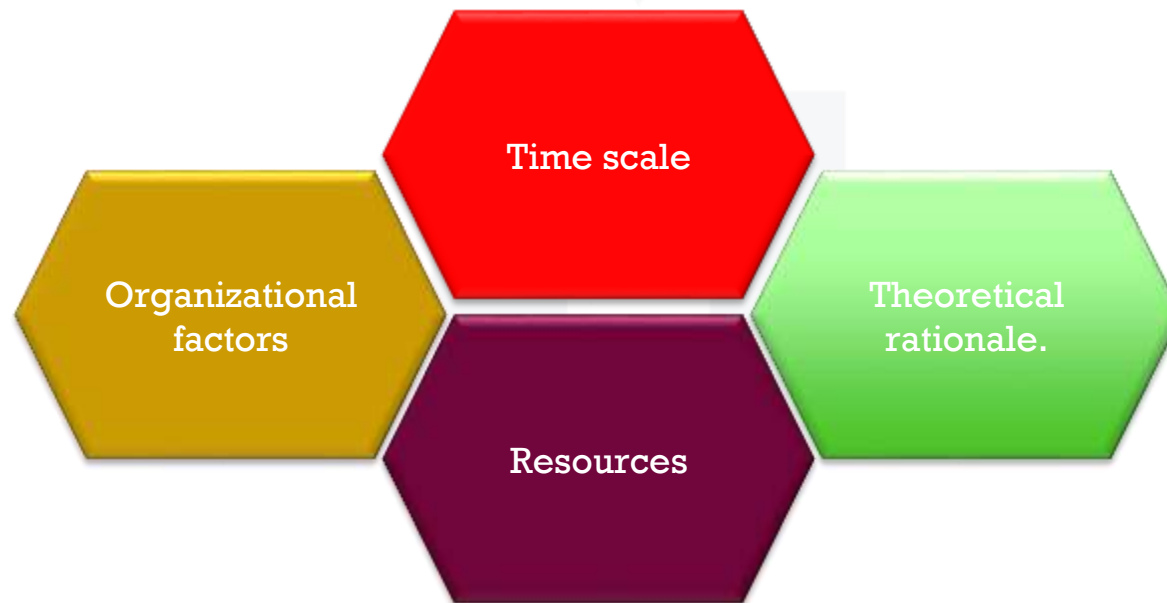


# Providing Guidance and Assistance

- Guidance can describe in
  - the form of the system's hard copy,
  - online documentation,
  - computer-based training,
  - instructional or prompting messages, and
  - system messages,
- Useful guidance and assistance answers the following questions:
  - What is this?
  - What does it do?
  - How do I make it do it?
  - What is its role in the overall scheme of things?

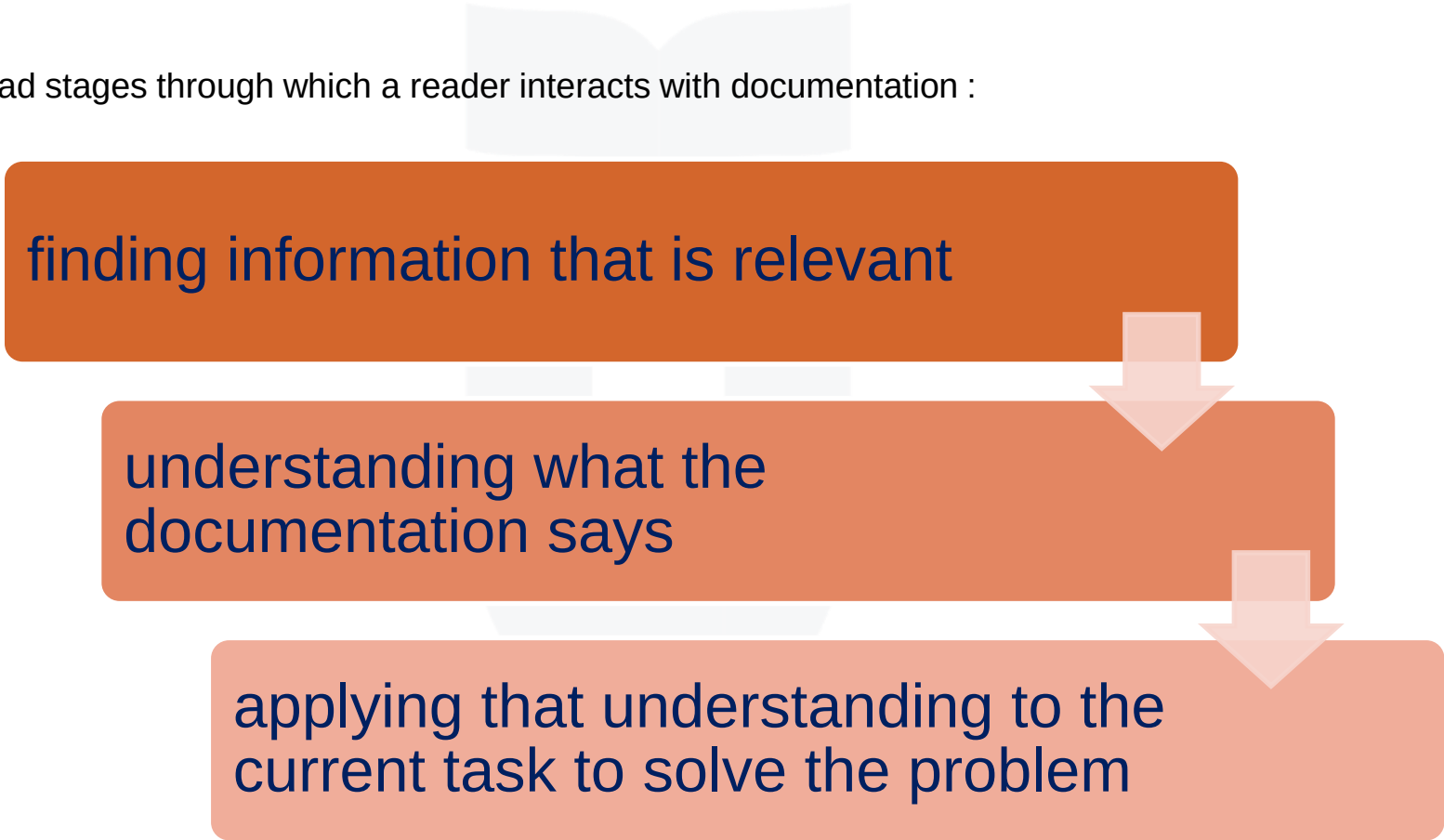
# Problems with Documentation

- factors contribute to bad design



# How Users Interact with Documentation

three broad stages through which a reader interacts with documentation :



finding information that is relevant

understanding what the  
documentation says

applying that understanding to the  
current task to solve the problem

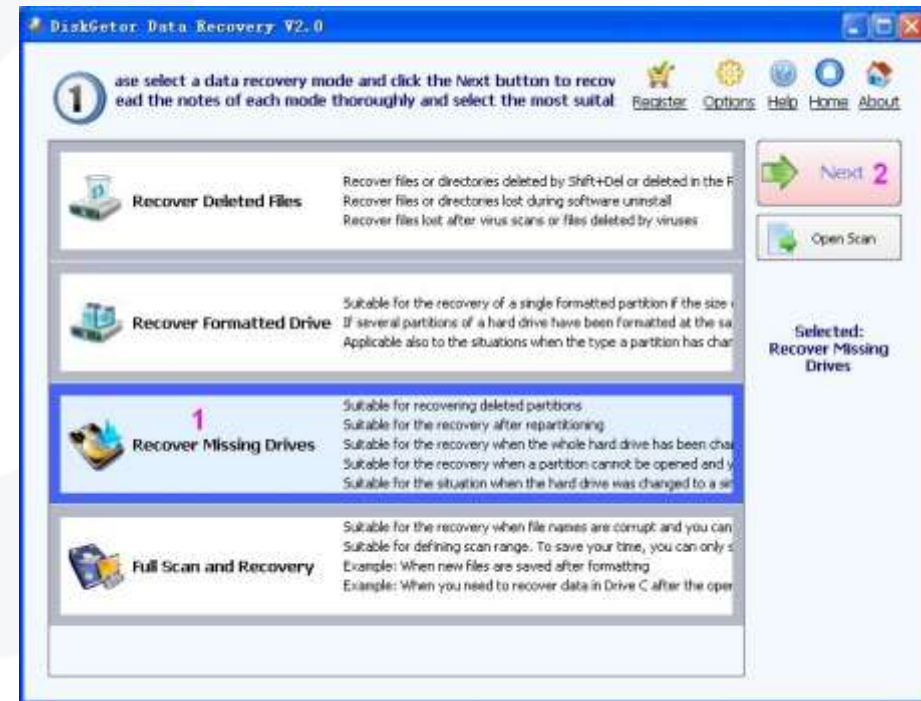
# How Users Interact with Documentation

- principles :



# Instructions or Prompting

- placed within the body of a screen
- Prompting is provided to assist a person in providing what is necessary to complete a screen
- Guidelines for writing instructions and prompts are covered in Step 8.



# Help Facility

## Help Facility Guidelines

- Kind:
- Training:
- Availability:
- Structure:
- Interaction:
- Location:
- Content:
- Style:
- Consistency:
- Title:



# Contextual Help

Provides information within the context of a task being performed

## *Help Command Button*

### ■ Description:

— A command button.



**Figure 9.2:** Help command button.

### ■ Purpose:

— To provide an overview of, summary assistance for, or explanatory information about the purpose or contents of a window being displayed.

### ■ Design guidelines:

— Present Help in a secondary window or dialog box.



# Contextual Help



**Figure 9.3:** Status bar message.

## *Status Bar Message*

- **Description:**
  - An abbreviated, context-sensitive message related to the screen item with the focus.
  - Appears in window's status bar when the primary mouse button is pressed over an item (or keyboard focus is achieved).
- **Purpose:**
  - To provide explanatory information about the object with the focus.

# Contextual Help

## *ToolTip*

- **Description:**
  - A small pop-up window that appears adjacent to control.
  - Presented when the pointer remains over a control a short period of time.
- **Purpose:**
  - Use to display the name of a control when the control has no text label.
- **Design guidelines:**
  - Make application-specific ToolTips consistent with system-supplied ToolTips.
  - Use system color setting for ToolTips above to distinguish them.



# Contextual Help

## ***What's This? Command***

- **Description:**
  - A command located on the Help drop-down menu on a primary window.
  - A button on the title bar of a secondary window.
  - A command on a pop-up menu for a specific object.
  - A button on a toolbar.
- **Purpose:**
  - Use to provide contextual information about any screen object.
- **Design guidelines:**
  - Phrase to answer the question “What is this?”
  - Indicate the action associated with the item.
  - Begin the description with a verb.

# Task-Oriented Help

## Description:

- A primary window typically accessed through the Help Topics browser.
- Includes a set of command buttons at the top; Purpose:
- To describe the procedural steps for carrying out a task.
- Focuses on how to do something.

## Design guidelines:

- Provide one procedure to complete a task, the simplest and most common.
- Provide an explanation of the task's goals and organizational structure at the start.
- Divide procedural instructions into small steps.
- Present each step in the order to be executed.

# Reference Help

## Description:

- An online reference book.
- Typically accessed through a
  - Command in a Help drop-down menu.
  - Toolbar button.

## Purpose:

- To present reference Help information, either

## Design guidelines:

- Provide a consistent presentation style, following all previously presented guidelines.
- Include a combination of contextual Help, and task-oriented Help, as necessary.
- Include text, graphics, animation, video, and audio effects,

# Wizards

## Description:

- A series of presentation pages displayed in a secondary window. — Include : Controls to collect input, Navigation command buttons.
- Typically accessed through

## Purpose:

- To perform a complex series of steps.
- To perform a task that requires making several critical decisions.
- To enter critical data and for use when the cost of errors is high.
- To perform an infrequently accomplished task.

# Hints or Tips

## Description:

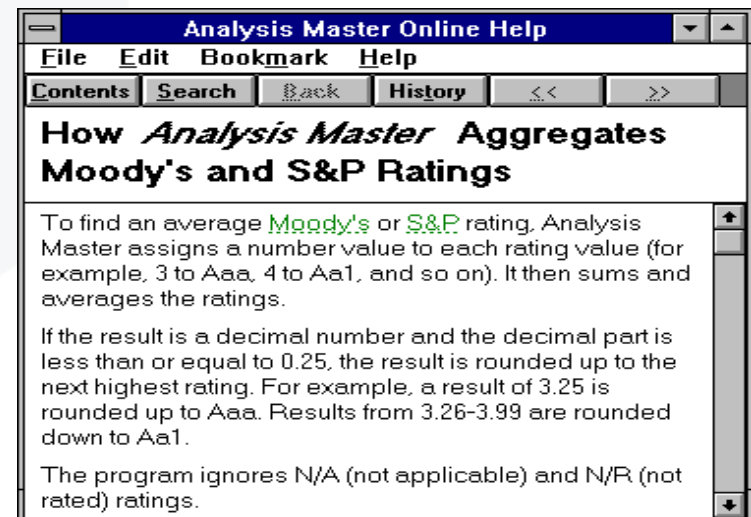
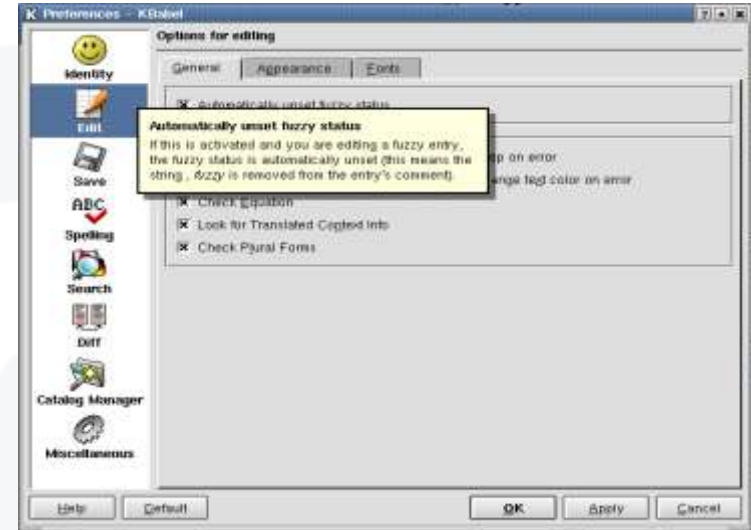
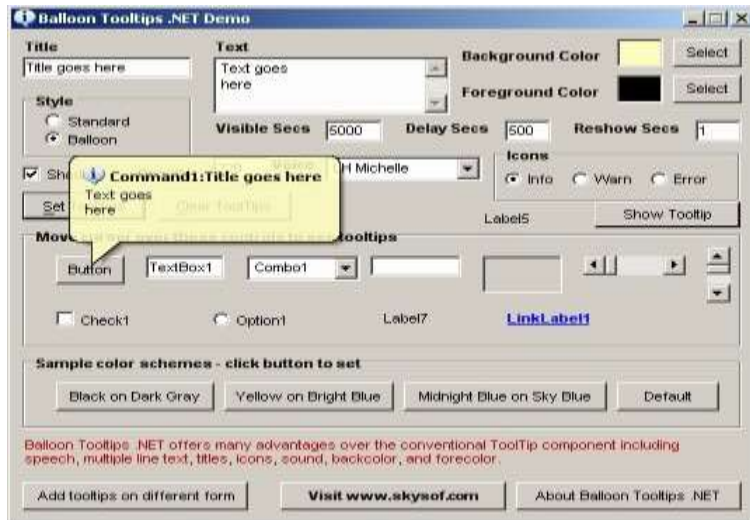
- A command button labeled Hints or Tips.

## Purpose:

- To provide a few important contextual, but specific, items of information related to a displayed screen

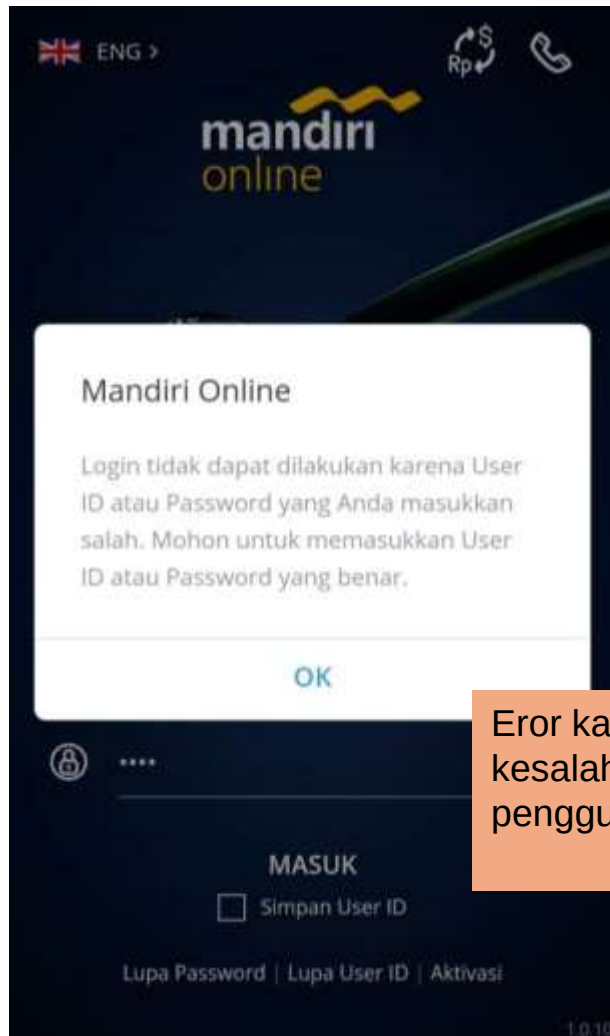
## Design guidelines:

- Provide guidance on only two or three important points.
- Locate the button near where its guidance applies.
- Write concisely and to the point

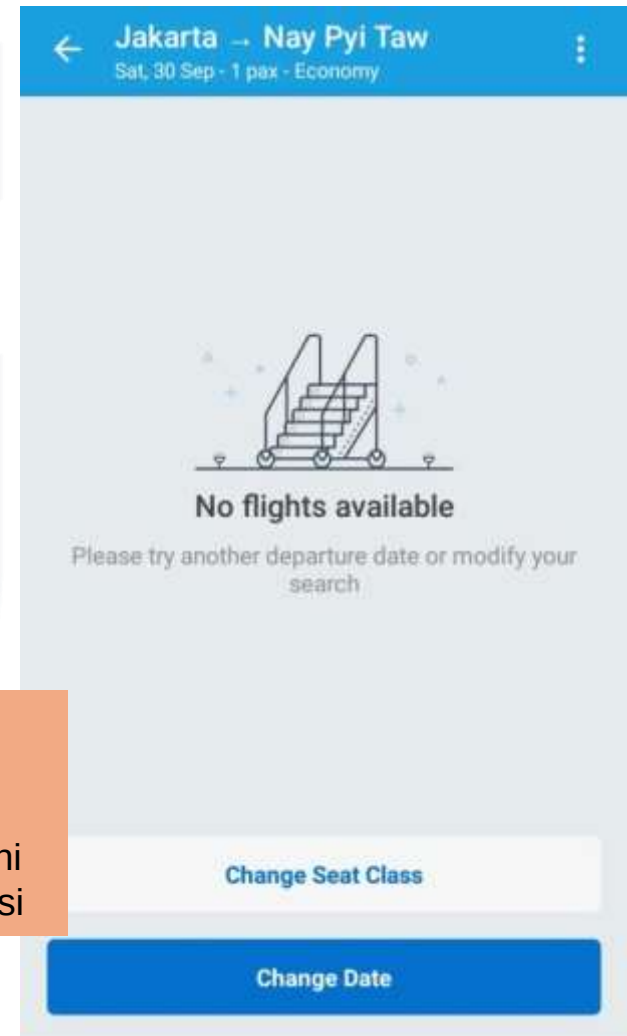




# Contoh studi kasus Guidances & Assistance



Eror karena kesalahan pengguna



Hasil operasi tidak memenuhi ekspektasi

# Activity

- Review carefully you project,
- Is there any feedback, guidance and assistance?

# Project Progress Presentation 1

- Prepare Your Paper prototype
- Prepare Your storyboard
- Prepare Your Report
- See U in Presentation 😊